Mini Motorways comparison

01 / Core gameplay and city logic

Current implementation, not planned features. Both playground versions refer to the current Unity app: OriginalReference and Custom. OriginalReference is not an exact replica.

Mechanic / feature	Actual game Standard play	Your playground OriginalReference	Your playground Custom	Open-source MiniMotorwaysRL
Maps	Full city maps; playable area expands.	Los Angeles, Beijing and Tokyo; fixed 50 x 30 crops.	Same three cropped city profiles.	Random 20 x 20 grid.
City growth	New houses and destinations appear over time.	Gradual spawning from recovered city data; approximate.	Same underlying growth.	Fixed initial buildings.
Color matching	Matching-color houses and destinations.	Houses serve matching-color destinations.	Same.	First matching business.
Household cars	Two reusable cars per house.	Two reusable cars per house.	Two reusable cars per house.	Five lifetime spawns per house.
Trip cycle	Destination visit and return home; repeated trips.	House > destination > original house; then reusable.	Same.	One-way; car disappears on arrival.
Destination demand	Demand accumulates; inadequate service can cause failure.	Demand accumulates; overcrowding pressure. Timing approximate.	Same, plus Custom grace and supply screening.	No destination-demand queue.
Destination upgrades	Larger destinations and more demanding layouts.	Missing; required for parity.	One size by design.	None.

Interpretation: the two playgrounds share traffic logic. Their main differences are construction resources, spending, grace periods and scenario rejection.

Mini Motorways comparison

02 / Traffic and infrastructure

Mechanic / feature	Actual game Standard play	Your playground OriginalReference	Your playground Custom	Open-source MiniMotorwaysRL
Routing	Automatic routing through roads and lanes.	Explicit road links and lane paths; exact parity unverified.	Same shared simulation.	Four-direction A* pathfinding.
Movement / congestion	Following, yielding and queues affect throughput.	Acceleration, following, queues and junction conflicts; approximate.	Same shared simulation.	Tile steps; overlap allowed.
Traffic lights	Functional intersection control.	Functional automatic signals; exact timing unverified.	Same.	Passable tile only; no signal logic.
Roundabouts	Functional circulation and yielding.	Directed circulation and entry handling; parity unverified.	Same.	Passable tile only; no circulation logic.
Parking / driveways	Dedicated parking and entrance layouts.	Parking, entrances and service flow implemented; layouts approximate.	Same.	Building tile is the endpoint.
Bridges	Limited upgrades for crossing water.	Water crossings use limited stock; geometry and accounting partial.	Water crossings add spending.	Passable tile; no water.
Motorways	Elevated connections bypass roads and obstacles.	Missing; required for parity.	Excluded by design.	Adjacent passable tile; no elevated bypass.
Demolition	Edits account for traffic already using infrastructure.	Retire road, clear committed traffic, release materials; timing unverified.	Same clearing; past spending is never refunded.	Immediate road / motorway removal.

Fidelity: functional traffic features do not establish exact original-game behavior. Motorways, destination upgrades, full maps and controlled trace comparisons remain key OriginalReference gaps.

Mini Motorways comparison

03 / Resources, experiments and features

Mechanic / feature	Actual game Standard play	Your playground OriginalReference	Your playground Custom	Open-source MiniMotorwaysRL
Construction limits	Limited materials in standard play.	Limited city-configured inventory.	Unlimited items; no active budget cap.	Fixed inventory.
Construction prices	Material availability; no Custom point-cost system.	No Custom point prices.	Road 2; light +5; bridge water tile 10; roundabout 8.	Resource counts.
Weekly upgrades	Periodic choices replenish roads and special items.	City-configured choices; partially reproduced.	None.	None.
New-shop grace	Original demand and failure rules.	Custom 30-second rule disabled.	30 seconds before failure / supply screening.	None.
Impossible scenarios	No equivalent Custom rejection mechanism.	Custom rejection disabled.	Approximate supply screen rejects clear shortages; demand is not lowered.	No screening.
Normal ending	Overcrowding in standard play.	Destination overcrowding.	Overcrowding; scenario rejection reported separately.	Step, stuck-car or resource thresholds.
Repeatability	Shared challenge scenarios; no project-style experiment interface.	Seeded map and event variation.	Same.	Clock-seeded randomness.
Learning / reward	Human gameplay; no built-in RL training interface.	No learner or scalar reward implemented.	Trips, time, waiting and spending tracked; weights undecided.	Random agent; no implemented Q-learning.
Programmatic access	No adapter from your project to the installed game.	Headless actions and observations; shared simulation core.	Same.	C++ observation vector.
Viewer / tools	Finished UI, sound, challenges and additional modes.	Unity viewer; inspection and lane-route display.	Same, plus spending information.	OpenGL visualization.

Scope: standard commercial gameplay is the main comparison. Creative and other modes may change the rules. The open-source column follows inspected code rather than unsupported README claims.

Reading the comparison

Scope, interpretation and sources

OriginalReference: an unfinished attempt to reconstruct the original game. Its goal is fidelity, but it is not yet an exact-game training baseline.

Custom: a deliberate change to the planning problem: unlimited construction, cumulative spending, a 30-second new-shop grace period and approximate supply screening. Survival is primary; waiting and spending have no chosen scalar reward weights.

MiniMotorwaysRL: the public KTK-Jadoo/mini_motorways_rl implementation inspected for this comparison. Lights, roundabouts and motorways are tile categories without their original traffic behavior. The included agent is random; the README describes capabilities that are not implemented in the inspected code.

Actual game: standard gameplay is the reference. Details about vehicle behavior also draw on the project's recorded read-only inspection of the installed game. Exact algorithms and timing have not been established by this comparison.

Project version specification - ../../simulation/Docs/VERSIONS.md

Project overview - ../../README.md

Behavioral fidelity notes - ../../simulation/Docs/FIDELITY.md

Open-source environment code

Open-source data structures

Open-source agent and application

Official game information